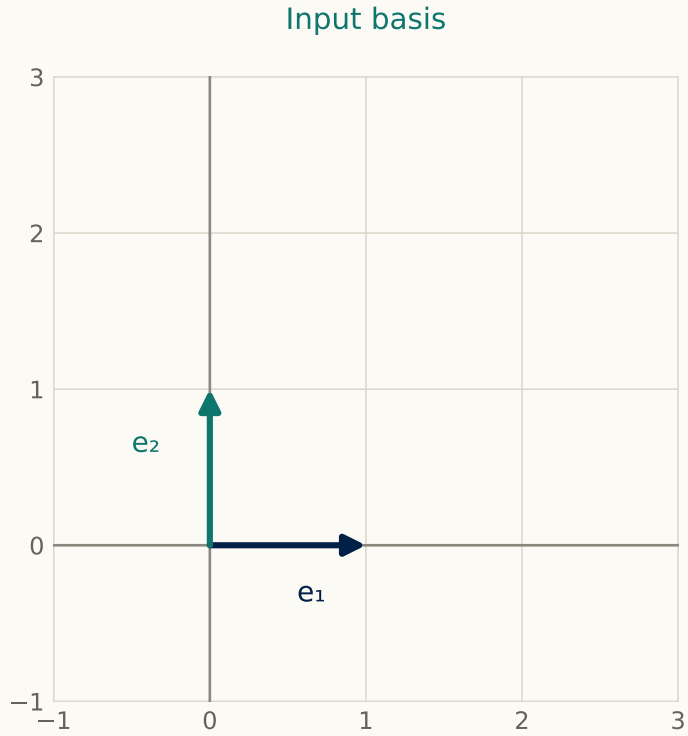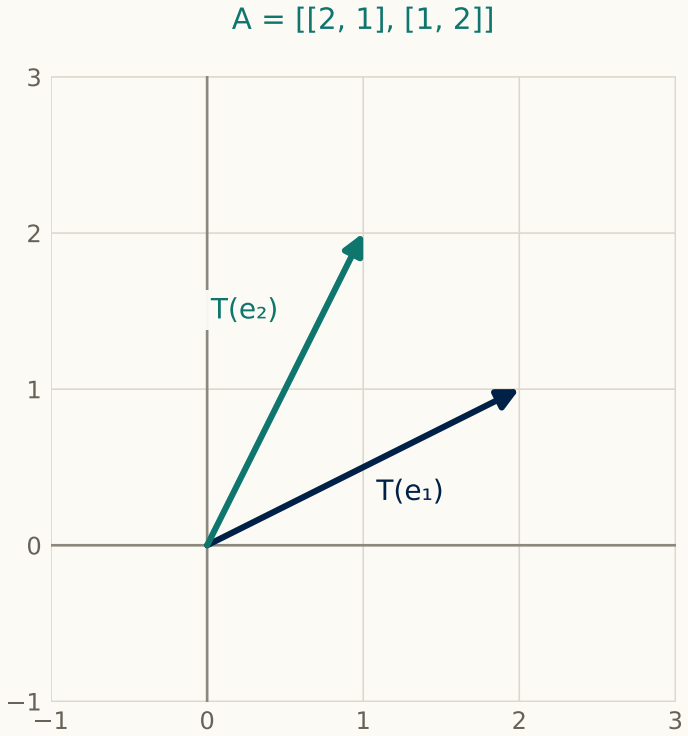The columns are images of the basis